

Skewness: Measuring the Shape of Data

Study notes. Descriptive statistics, measures of asymmetry. Built from the class notes (Part 1), the course slides and the skewness example workbook.

The whole idea in three lines

1. **Skewness** tells you whether data is balanced around its centre or piled up on one side.
2. A skew is named after the side of its **long tail**, not the side where most values sit.
3. The **mean** is dragged toward the tail while the **median** stays put, so comparing them reveals the skew.

1. Where skewness fits

Descriptive statistics describes a single variable (a *univariate* description) in three complementary ways:

Family	Question it answers	Measures
Central tendency	Where is the centre?	Mean, median, mode
Asymmetry	Is the data balanced or lopsided?	Skewness
Variability	How spread out is it?	Variance, standard deviation, coefficient of variation

You read skewness *using* the mean, median and mode, which is why it comes right after them in the course.

2. What is skewness?

Skewness is the measure of asymmetry. It indicates whether the observations in a dataset are concentrated on one side. A perfectly symmetric dataset is a mirror image around its centre. When a few values stretch far out on one side, the picture becomes lopsided, and skewness measures by how much and in which direction.

3. The three types

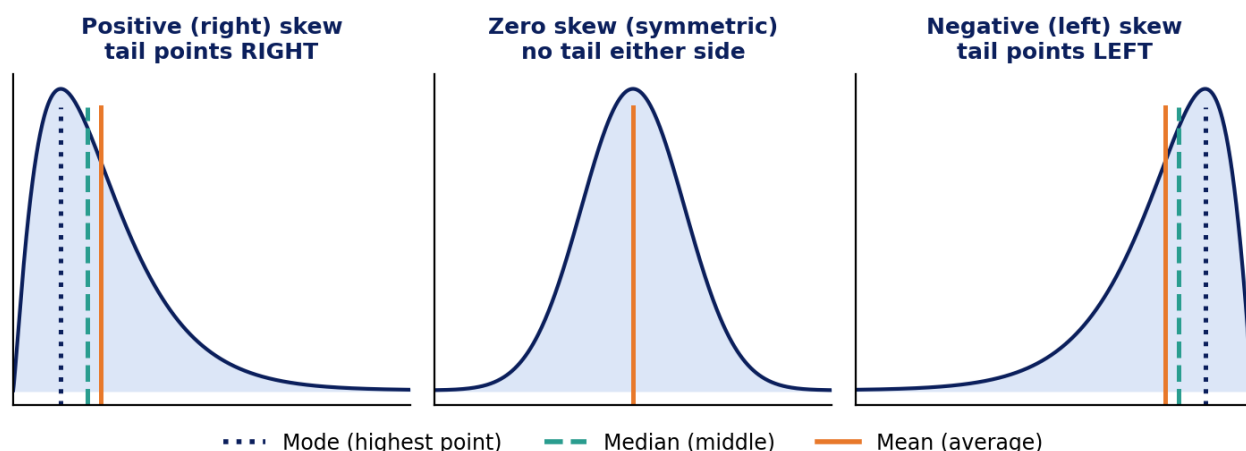


Figure 1. The three shapes (illustration). Notice how the three averages change order as the tail changes side.

	Positive (right) skew	Zero skew (symmetric)	Negative (left) skew
Long tail points	Right	Neither side	Left
Most values sit	On the left (lower values)	Evenly around the centre	On the right (higher values)
Order of the averages	Mode < median < mean	Mean = median = mode	Mean < median < mode
Sign of skewness	Positive (+)	Zero (0)	Negative (-)
Typical example	Incomes, house prices	Adult heights (roughly)	Scores on an easy exam

The typical examples are common illustrations, not data from your class.

Remember: the tail tells the tale

The name follows the **tail**. In a positive (right) skew, the *bulk* of the data is on the left, but the *tail* (the outliers or extreme values) stretches to the right. It is a common slip to read 'right skew' as 'data on the right'. It means the opposite.

A second memory hook: **the mean chases the tail**. It always sits on the tail side of the median.

4. Why the mean moves but the median does not

The mean uses **every** value, so a few extreme values pull it toward them. The median is just the middle position of the ordered data, so extreme values barely affect it. When the two disagree, something is dragging the mean, and that something is a long tail.

In Dataset 1 below, most values are small (1 to 3), but a single value of 7 stretches the right tail. It lifts the mean to **2.79** while the median stays at **2**.

5. Worked example: three datasets

The example workbook compares three datasets of about 20 values each. Each is summarised below, with its mean, median and mode drawn on top.

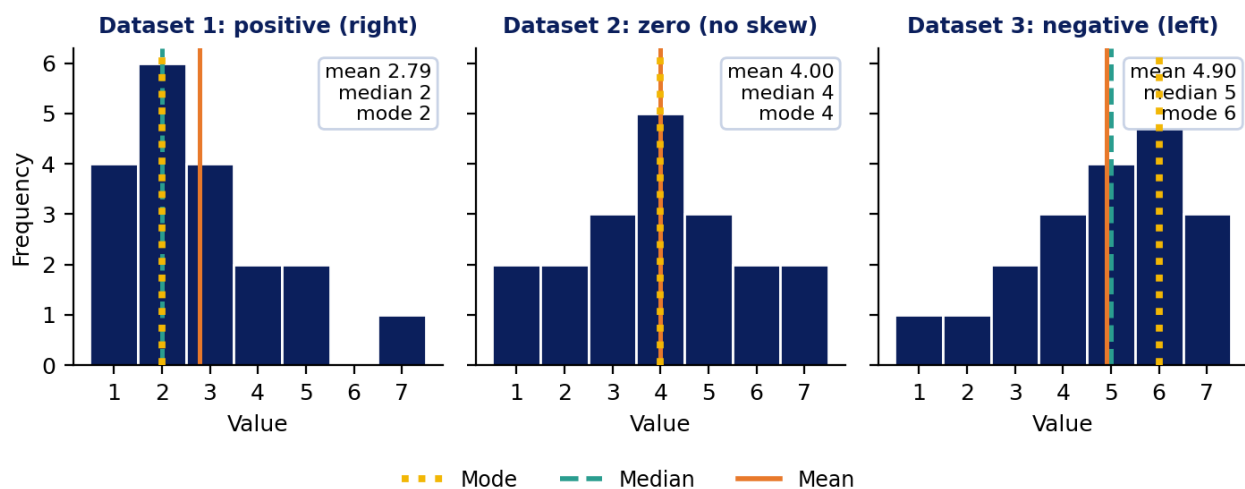


Figure 2. The workbook's three datasets (bars = how often each value occurs). In Dataset 1 the median and mode coincide at 2.

	Dataset 1	Dataset 2	Dataset 3
Number of values	19	19	20
Mean	2.79	4.00	4.90
Median	2.00	4.00	5.00
Mode	2.00	4.00	6.00
Mean compared with median	Mean > median	Mean = median	Mean < median
Verdict	Positive (right) skew	Zero skew	Negative (left) skew
Excel =SKEW()	+1.08	0.00	-0.78

How to diagnose skew in four steps

1. Calculate the mean, median and mode. 2. Compare the mean with the median: higher means a right skew, lower means a left skew, equal means symmetric. 3. Look at the histogram to see which side the tail is on. 4. Confirm with a number: =SKEW(range) in Excel.

6. Measuring it with a number

The class slides give the sample skewness formula:

$$\text{Skewness} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left(\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^{3/2}}$$

Figure 3. Sample skewness formula.

In plain words:

1. For each value, subtract the mean to see how far it sits from the centre.
2. **Cube** each difference. Cubing keeps the sign, so values above the mean give positive numbers and values below give negative ones. It also makes big differences dominate, so a long tail decides the result.
3. Divide by the spread (standard deviation) cubed, so the answer has no units and can be compared across datasets.

Result	Meaning
Above 0	Positive (right) skew
Around 0	Symmetric
Below 0	Negative (left) skew

In practice you let software do it: =SKEW(range) in Excel. Excel uses a slightly adjusted version of the formula, so its numbers differ a little from the slide formula (for Dataset 1: 1.08 in Excel, 0.92 from the slide formula). The sign and the interpretation are the same.

Rule of thumb for the size

A commonly used guide, not part of your class notes: skewness between -0.5 and +0.5 is roughly symmetric, between 0.5 and 1 (either sign) is moderately skewed, and beyond 1 is highly skewed.

7. Why skewness matters

Reason	In practice
It tells you where the data sits	Skewness shows whether values bunch up low or high, which a single average hides
It guides which average to report	With a strong skew the mean is pulled by the tail, so the median is often the better 'typical' value
Averages work best together	Use mean, median and mode side by side for a fuller picture of the data
It flags a lopsided variable early	Later, in data analysis and machine learning, skewed variables are often transformed before modelling

8. Common mistakes

Mistake	Correct idea
Thinking 'right skew' means the data is on the right	The tail is on the right; most data is on the left
Naming the skew by where the peak is	Name it by where the tail points
Treating mean > median as proof of positive skew	It is a quick check that usually works. Confirm with a histogram or =SKEW()
Looking at the mean alone	Compare mean, median and mode together

9. Quick check

#	Question	Answer
1	A dataset has mean 60 and median 52. What kind of skew?	Positive (right): the mean is above the median, tail on the right
2	Mean, median and mode are all equal. What does that tell you?	Symmetric, zero skew
3	In a left-skewed dataset, which is larger, the mean or the median? Where is the tail?	The median is larger. The tail is on the left
4	True or false: in a positively skewed dataset most values are on the right.	False. Most are on the left; the tail is on the right

Summary

1. Skewness measures asymmetry: is the data piled on one side?
2. **Positive (right):** tail right, mode < median < mean. **Zero:** mean = median = mode. **Negative (left):** tail left, mean < median < mode.
3. Name the skew by the tail, and remember that the mean chases the tail.
4. Diagnose with mean vs median, a histogram, and =SKEW() in Excel.